

The following Karnaugh map represents a function F .

		YZ			
		00	01	11	10
X	0	1	1	1	0
	1	0	0	1	0

Q.52 A minimized form of the function F is

- (A) $\overline{X}Y + YZ$ (B) $\overline{X}\overline{Y} + Y\overline{Z}$
 ✓ (B) $\overline{X}\overline{Y} + YZ$ (C) $\overline{X}\overline{Y} + \overline{Y}Z$

Solution:

From the K-map,

- The two adjacent 1's in the first row corresponding to $YZ = 00$ and $YZ = 01$ are grouped together.

Here, $X = 0$ and $Y = 0$ remain constant.

Hence, the term obtained is

$$\overline{X}\overline{Y}$$

- The two 1's in the column corresponding to $YZ = 11$ are grouped together.

Here, $Y = 1$ and $Z = 1$ remain constant.

Hence, the term obtained is

$$YZ$$

Therefore,

$$F = \overline{X}\overline{Y} + YZ$$

Hence, the correct answer is

(B)